

Lab 5: Pseudoscience

Answer Key

“Anti-intellectualism has been a constant thread winding its way through our political and cultural life, nurtured by the false notion that democracy means that ‘my ignorance is just as good as your knowledge.’” — Isaac Asimov

1 Introduction

What is science? What is pseudoscience? As you navigate today’s world, rife with claims stemming from junk science (or no science at all), it is important to be able to discern between the two. In today’s lab, we will give you the qualitative tools to make the distinction between science and pseudoscience. We will also disprove some pseudo-scientific theories, using probabilities and geometry.

2 What Makes Something Scientific?

A theory, model, prediction, measurement, or observation is generally considered **scientific** if it is:

- (a) **Repeatable** (at least in principle, as may be the case with rare events we cannot control)
- (b) **Objective**
- (c) **Predictive**
- (d) **Falsifiable**

1. *[4 points]* Briefly, what do each of these four words mean to you? Discuss with your group, and write down your answers.

Answer: 1 point for each word explained. Answers just need to get the gist of it:

Repeatable: an experiment can be done over and over again, and always give you the same result.

Objective: Outcome doesn’t change depending on who/what does it

Predictive: there’s an idea of what will happen before it happens.

Falsifiable: there exists a possible observation that would prove the claim wrong

What constitutes a scientific theory? A theory is an explanation of some aspect of the natural world that has been repeatedly upheld by rigorous experiment and never falsified. Theories begin as hypotheses: testable predictions that can be corroborated or falsified via the scientific method. Theories are similar to scientific laws, but are usually broader in scope (their explanatory power extends further). For instance, Kepler's First Law states that the orbit of every planet is an ellipse with the Sun at one focus. Clearly, this describes a specific phenomenon, in contrast to, for instance, the Theory of General Relativity, which explains all phenomena associated with gravitation. Both laws and theories are scientific fact. One common misconception is that scientific theories are “just theories”: that they have yet to become scientific law, or that they are on equal footing with one or more alternative ideas.

3 Pseudoscience

Pseudoscience refers to any belief, claim, or practice that is presented as scientific but does not actually follow the scientific method or possess the characteristics described in Section 2. Pseudoscience often uses the language or appearance of science without following the standards that make a claim scientifically testable. Examples of pseudoscience range from the fairly harmless (cryptozoology – the belief that creatures like the Loch Ness monster exist) to the actively dangerous (medical pseudoscience that leads people to forgo real treatment) to the historically horrific (“scientific racism,” the discredited claim that some races are biologically superior to others). It is also important to distinguish between a claim that is wrong, a claim that has not been tested, and a claim that is presented as scientific but does not actually allow itself to be tested. Plenty of real science turns out to be wrong. And “not yet tested” is a *status*: an untested claim is one somebody could go and check tomorrow. A pseudoscientific claim is one where attempting to verify it reveals that it is incompatible with the scientific method at all.

4 Evaluating Scientific Claims

Let's practice using the above 4 criteria on some claims. Scientists must think critically about the information they consume, whether it is found on Instagram reels or in a paper on the arxiv. Below are three claims, in no particular order.

- A1.** Mercury is in retrograde, so you should avoid signing contracts this week.
- A2.** People who sleep less than six hours a night are more likely to report symptoms of depression.
- A3.** People who drink coffee live longer.

2. Let's evaluate claim **A1** using the criteria from Section 2.

- (a) [1 points] What observation or result, if any, would convince you that this claim is *false*? Consider the non-committal language: “should avoid”.

Answer: Nothing. The way A1 is phrased means it can be waved off as a fluke if someone signs a contract during Mercury retrograde, and it goes well.

- (b) [2 points] Would you classify this claim as **scientifically testable**, **pseudoscientific**, or **not yet tested**? From the four criteria in Section 2, choose *one* and explain why it cannot be met.

Answer: Pseudoscientific. It isn't objective (a "bad" outcome is subjective), it isn't predictive (it doesn't say anything about the type of contract or how much risk to expect etc.), and it isn't falsifiable (any bad outcome "confirms" it, any good outcome can be explained away - this relates to the previous part).

- (c) [1 points] Suppose astrology has no real effect on how contracts turn out. Why might someone still be convinced, from their own experience, that the advice works?

Answer: One point each for anything reasonable. Examples:

- If you've been told a week is risky, you will notice and remember the mishaps more than anything that went fine (confirmation bias).
- "Goes wrong" is vague enough that almost any week qualifies.
- If you think the week will go badly, you may unknowingly rush, or second-guess decisions and get a worse outcome. A self-fulfilling prophecy!
- If you've got friends who really believe in astrology, they might reinforce the belief.

It is tempting to jump from a correlation straight to cause and effect: since Y followed X, X *must* have caused it. However, correlation doesn't prove causation. X might cause Y, Y might cause X, or a hidden third factor could be driving both, or it could simply be a coincidence. While correlations are great for sparking curiosity and guiding further research, we must be careful not to treat them as anything more.

3. [2 points] Claims **A2** and **A3** are both based on correlations found in real data. For each, give one possible explanation other than "X causes Y."

Answer: 1 point each for a reasonable alternative explanation.

A2 (sleep/depression): depression itself disrupts sleep. An external factor like stress or illness can drive both short sleep and low mood.

A3 (coffee/longevity): Someone healthy enough to make it to old age would have more time to drink more coffee (or something like that). People who drink coffee may have higher incomes (coffee is more expensive than tea), or be more active, or be more social, which can increase longevity.

5 Evaluating an Argument

Let's look at a confident, AI-generated (because this is the world we live in) argument, where it takes a little more discernment to evaluate its credibility. The questions below help us evaluate how well a particular argument is actually constructed.

“Hospital emergency rooms and police stations consistently report a dramatic surge in psychiatric admissions, violent crimes, and trauma cases during the full moon. Because the human body is composed of roughly 60% water, the Moon’s gravitational pull exerts a tidal force on human brain tissue and bodily fluids in the exact same way it creates ocean tides. This monthly gravitational stress disrupts neurological activity and sleep cycles, directly causing heightened aggression and erratic behavior across human populations during full lunar phases.”

4. Answer the following questions in 1-2 sentences each:

- (a) [1 points] The argument asserts that emergency rooms and police stations “consistently report a dramatic surge” during full moons. Before trying to explain *why* this happens, what step must a scientist take regarding this initial premise?

Answer: A scientist must first check whether the claimed effect actually exists in empirical data (i.e., check if the premise is true).

- (b) [1 points] The author claims that lunar gravity exerts a “tidal force on human brain tissue” because humans are made of water. What is flawed about this argument?

Answer: Tidal forces depend on the size of the object across which gravity acts. Humans are very tiny compared to the ocean, so it has a very negligible effect on us.

- (c) [1 points] The argument claims that full moons cause increased aggression and erratic behavior. Give one possible alternative explanation for the claimed pattern.

Answer: 1 point for identifying relevant evidence, 1 point for a reasonable alternative explanation, and 1 point for explaining how the evidence could distinguish between them. Examples of alternative explanations include coincidence, differences in reporting behavior, or other factors that happen to coincide with full moons.

6 Astrology & the Binomial Distribution

Astrology proposes that the position of celestial objects (e.g. the sun, moon, and planets) influences terrestrial events. It is rooted in ancient practices from all over the globe, with signs of it being practiced from the 2nd millennium BCE! It’s a very human desire to understand why certain events happen (good or bad), and it’s remarkable how well-studied the night sky became because of it. Indeed, there was almost no distinction between astrology and astronomy until the late Middle Ages.

These days, we may be exposed to astrology through social media or apps that generate **horoscopes**: regular predictions aligned with ones astrological natal chart. These horoscopes give advice like “do something that scares you today” or “avoid making decisions about your living situation.” Horoscopes are fun to check, but it is language that could arguably apply to anyone ¹.

¹This is such a common cognitive bias, that it has a name: the Barnum effect.

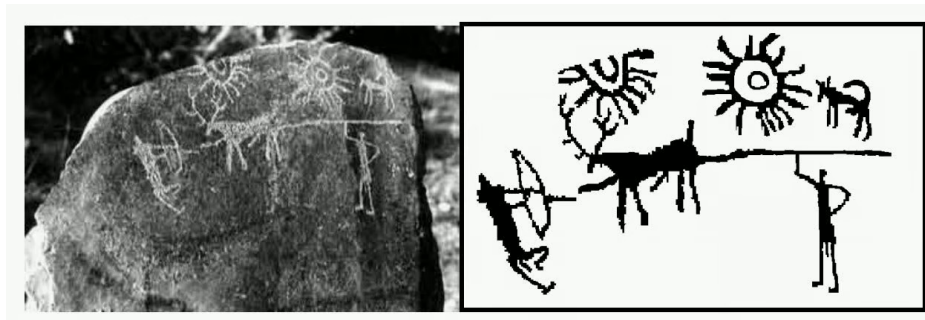


Figure 1: Rock art dated to roughly 2100 BCE, that some astronomers believe depict the supernova HB9. Source: Quartz.com

Let's check horoscopes for the Sun signs from three different sources (see handout), where the signs have been stripped off. From each of the three sources, pick the one that best matches your week so far. Once everyone has chosen, we'll reveal the actual signs and see how you did. For reference:

Aries: Mar 21 – Apr 19	Libra: Sep 23 – Oct 22
Taurus: Apr 20 – May 20	Scorpio: Oct 23 – Nov 21
Gemini: May 21 – Jun 20	Sagittarius: Nov 22 – Dec 19
Cancer: Jun 21 – Jul 22	Capricorn: Dec 22 – Jan 19
Leo: Jul 23 – Aug 22	Aquarius: Jan 20 – Feb 18
Virgo: Aug 23 – Sep 22	Pisces: Feb 19 – Mar 20

6.1 The Binomial Distribution

If astrology worked, everyone would pick the horoscope matching their real sign every time. If it doesn't work, that doesn't mean nobody ever picks correctly – pure chance will get some people right some of the time. If the horoscope choices are unrelated to a person's actual sign, each choice has a probability $p = 1/12$ of being correct. Treating the choices as independent, the **binomial distribution** gives the probability of getting exactly k correct out of n attempts:

$$P(k \mid n, p) = \frac{n!}{k! (n - k)!} p^k (1 - p)^{(n - k)}. \quad (1)$$

5. [3 points] What are p , k , and n for our horoscope experiment?

Answer: $n = 3$ is the number of horoscope sources each student picks from (one guess per source). $p = 1/12$ is the probability of guessing any one correctly. k is the number of correct guesses out of the 3 — this varies!!!

6. [4 points] What is the probability of guessing zero, one, two, or three horoscopes correctly by pure chance?

Answer: Probabilities (with $n = 3$, $p = 1/12$): $P(0) = (11/12)^3 \approx 0.770$; $P(1) = 3(1/12)(11/12)^2 \approx 0.210$; $P(2) = 3(1/12)^2(11/12) \approx 0.019$; $P(3) = (1/12)^3 \approx 0.0006$. (quick check: these four values should sum to ≈ 1)

7. [2 points] Plot the four results from above: $P(k | n, p)$ on the y -axis, k on the x -axis. Label the axes.

Answer: 1 point for correct labels. 1 point for all 4 points.

8. [3 points] How many did you get right? Take a few minutes to compare with your group-mates/classmates. Add their k -values to your plot, in a different symbol. Does the spread of correct guesses roughly follow the binomial distribution?

Answer: 1 point for stating their k . 1 point for including their classmates values in the plot. 1 point for evaluating the spread.

9. [1 points] Based on tonight's results, what do you conclude about the predictive power of horoscopes?

Answer: Not predictive (unless for some reason everyone guesses really well...

7 Flat Earth

The Flat Earth theory is self-explanatory: it is the idea that the Earth is shaped like a flat plane, rather than a sphere. This claim is useful to examine because it makes a clear, testable prediction: if the Earth is spherical, distant objects should eventually disappear below the horizon. In this section you'll test that prediction quantitatively, using two methods built on the same right triangle.

7.1 How Far Can You See?

If the Earth were flat, you could see infinitely far, limited only by haze and the resolving power of your eyes or telescope. On a spherical Earth, the surface curves away from you and you are eventually limited by the **horizon distance**. For an observer at height h above the surface of a sphere of radius R , the distance to the horizon is

$$d = \sqrt{2Rh + h^2} \approx \sqrt{2Rh} \quad (2)$$

where the approximation holds when $h \ll R$. This is an application of the Pythagorean theorem.

10. The Earth's radius is 6,371 km. Be careful about units.

- (a) [2 points] A person standing at the shore with their eyes 1.7 m above sea level can see how far to the horizon?

Answer: (a) $h = 1.7 \text{ m} = 0.0017 \text{ km}$: $d = \sqrt{2(6371)(0.0017)} \text{ km} \approx \sqrt{21.66} \approx 4.65 \text{ km}$.

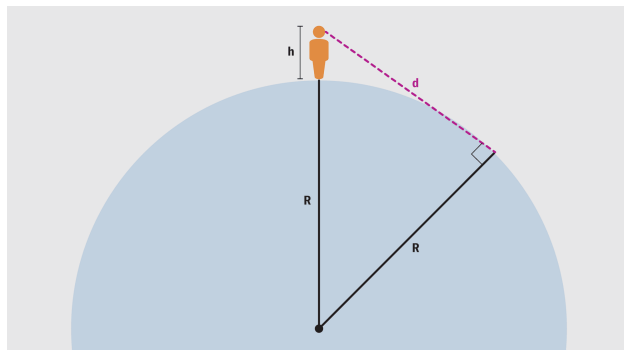


Figure 2: Visual of the horizon distance, d . Source: Scientific American

- (b) [2 points] The Top of the Rock observation deck in Midtown Manhattan is about 260 m above sea level. How far can you see from there on a clear day?

Answer: $h = 260 \text{ m} = 0.260 \text{ km}$: $d = \sqrt{2(6371)(0.260)} \text{ km} \approx \sqrt{3312.9} \approx 57.5 \text{ km}$.

- (c) [2 points] If the Earth were flat, would either of those numbers exist at all? Explain.

Answer:

(c) No. On a flat Earth there is no geometric horizon.

11. [3 points] Flat Earth proponents may point to photos or videos where a distant city skyline is visible across a lake as “proof” that there’s no curvature. Using your horizon-distance formula, explain why tall buildings can remain visible well beyond the point where the shoreline itself has disappeared below the horizon.

Hint: compare h for the shoreline and for a tall building. What changes with h ?

Answer: Horizon distance grows with $\sqrt{h}$. The shoreline itself has $h \approx 0$, so its own horizon distance is essentially zero: it disappears from view almost immediately. A tall building has a much larger h , so $d = \sqrt{2Rh}$ is correspondingly larger — it stays visible.

7.2 Al-Biruni’s Measurement

Around 1020 CE, the Persian polymath Abū Rayhān al-Bīrūnī stood on a mountain near the Fort of Nandana, in what is now Punjab, Pakistan. He wanted to estimate the Earth’s radius without already knowing its value.

Figure 2 shows the geometry of the measurement. The observer is a height h above the Earth’s surface, and the line of sight to the horizon is tangent to the Earth. The radius drawn to the horizon is therefore perpendicular to the line of sight.

12. [2 points] The **dip angle** θ is the angle between the observer’s true horizontal and their line of sight to the horizon (d). Which angle in the right triangle is equal to θ ? (It might be helpful to sketch it out)

Answer: θ is the angle made between $R+h$ and R . a.k.a the angle at Earth's center between the radius to the observer and the radius to the horizon point.

13. [3 points] Write an equation relating θ , R , and h . Then solve your equation for R .

Answer: 1 point for writing the right equation. 1 point for good set-up and work shown. 1 point for getting the right equation for R .

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{R}{R+h}. \text{ Solving for } R: \cos \theta (R+h) = R \Rightarrow R \cos \theta + h \cos \theta = R \Rightarrow$$

$$h \cos \theta = R - R \cos \theta = R(1 - \cos \theta) \Rightarrow R = \frac{h \cos \theta}{1 - \cos \theta}.$$

14. [2 points] Al-Bīrūnī measured a mountain height of roughly 305 m and a dip angle of about 0.57° . Use your equation from the previous part to estimate the Earth's radius. Report your answer in km. How close is it to the true answer, percentage-wise?

$$R_{\oplus} = 6371 \text{ km.}$$

Answer: 1 point for correctly plugging-and-chugging. 1 point for comparing in terms of percentage (it's okay if this is a little off, but should be around 3%).

For $\theta = 0.57^\circ$:

$$\cos(0.57^\circ) \approx 0.999951 \quad \Rightarrow \quad 1 - \cos(0.57^\circ) \approx 0.0000495$$

$$R = \frac{305 \times 0.999951}{0.0000495} \approx 6,163,000 \text{ m} \approx \mathbf{6,163 \text{ km}}$$

. This is within about 3% of the accepted 6,371 km.

15. [1 points] Suppose Al-Bīrūnī had climbed to a higher mountain. Would you expect the dip angle to be larger or smaller?

Answer: Larger. From $\cos \theta = R/(R+h)$, as h increases the denominator $R+h$ grows while R stays fixed, so $\cos \theta$ decreases.

8 Reflection

16. [1 points] Take a few minutes to reflect on what you learned today. Have you encountered examples of pseudoscience in real life e.g. through social media or word of mouth?

Answer: Free point as long as they answer